

Stanford University
Political Science 350D
Problem Set D
Due April 6, 2015

1. In a study of teen recidivism, there were $n = 43$ persons released from incarceration, of whom $y = 15$ were re-offenders within 36 months. We wish to estimate the probability θ of recidivism.
- (a) Using a Uniform(0, 1) prior for θ , plot the prior, likelihood, and posterior distributions of θ . Find the posterior mean, median, mode and standard deviation of θ and a 95% credible interval for θ .
 - (b) Repeat (a), but use a Beta(2, 8) prior for θ .
 - (c) Repeat (a), but use a Beta(8, 2) prior for θ .
 - (d) Consider the following prior distribution for θ :

$$p(\theta) = \frac{1}{4} \frac{\Gamma(10)}{\Gamma(2)\Gamma(8)} (3\theta(1-\theta)^7 + \theta^7(1-\theta)),$$

which is a 75-25 percent mixture of a Beta(2, 8) and a Beta(8, 2) prior distribution. Plot this prior distribution and compare it to the priors in (b) and (c). Describe what sort of prior opinion this may represent.

- (e) For the prior in part (d):
 - i. Write an expression for $p(\theta)p(y|\theta)$ and simplify as much as possible.
 - ii. The posterior distribution is a mixture of two distributions you should know. Identify these distributions from their kernels.
 - iii. Use R to calculate and plot the unnormalized posterior $p(\theta)p(y|\theta)$ by evaluating it on a grid of points. Identify the (approximate) posterior mode and compare to your answers in parts (a–c).
 - (f) Find a general formula for the weights on the mixture distribution in (d)(ii) and provide an interpretation for these values.
2. Install and use JAGS to answer Question 1(b).